



Society of Petroleum Engineers

SPE-229065-MS

Investigating the Global Potential of Hydrogen Technology: Emerging Trends, Investments, and Regional Comparisons

Y. H. Ismail and D. E. Sintim-Aboagye, Siemens Energy, Abu Dhabi, Abu Dhabi, United Arab Emirates

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229065-MS](https://doi.org/10.2118/229065-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

As hydrogen gains momentum as a cornerstone of the global energy transition, countries are racing to define their roles in the emerging hydrogen economy. This paper investigates the question: Who will lead the hydrogen market in the future? Through a comparative analysis of national hydrogen strategies, investment volumes, project maturity, and regulatory frameworks, the study identifies significant regional disparities in ambition, feasibility, and execution. A novel application of the Growth Rate of Hydrogen Capacity is used as a predictive indicator to assess the annual production increases required to meet national targets. The analysis also explores the correlation between investment and project completion, revealing that financial commitment alone does not guarantee progress. While Europe and the Americas currently lead in project maturity and regulatory readiness, the Middle East and Asia Pacific are emerging as high-investment regions with long-term export potential. Africa, though still in early stages, shows promise as a future green hydrogen supplier. The findings offer a multidimensional view of strategic readiness and highlight the critical role of policy alignment, infrastructure development, and international cooperation in shaping hydrogen market leadership.

Introduction

Hydrogen is no longer viewed solely as a fuel alternative, it is increasingly regarded as a fundamental pillar of global energy and economic systems, central to nearly every region's net-zero emissions strategy. Its capacity to decarbonize carbon-intensive sectors such as steel, chemicals, and long-haul transport, combined with its compatibility with renewable energy systems, has positioned hydrogen as a cornerstone of the global energy transition particularly due to its versatility and potential to decarbonize sectors that are hard to electrify.

This paper seeks to answer a critical question: *Who will lead the hydrogen market in the future?* A question that resonates with policymakers, investors, and analysts alike. To address this, the study conducts a comparative analysis of national hydrogen strategies, investment volumes, project maturity, and regulatory frameworks across global regions. The objective is not only to assess current progress but to forecast strategic readiness and identify the most versatile leadership potentials in the evolving hydrogen economy.

Methodology

The methodology draws on a diverse set of quantitative sources, including the International Energy Agency (IEA, 2024a), the International Renewable Energy Agency (IRENA, 2024), published national hydrogen strategies, and regional project databases. Several analytical tools are employed:

- **Required Growth Rate (RGR) of Hydrogen Capacity:** Repurposed in this study as a predictive indicator, this metric estimates the annual increase in hydrogen production required for each country or region to meet its stated targets and measures how fast a region should go in scaling up their hydrogen infrastructure (Corporate Finance Institute, n.d.).

$$RGR = \left(\frac{\text{Final Target Hydrogen Annual Production}}{\text{Current Hydrogen Annual Production}} \right)^{1/N} - 1 \quad (1)$$

N is the number of years between current and the final targeted hydrogen annual production. This approach enables a forward-looking assessment of strategic feasibility.

- **Correlation Between Investment and Project Completion (r):** Using the Pearson correlation coefficient, this metric evaluates whether higher investment levels are associated with increased project delivery, offering insights into capital efficiency (Benesty et al., 2009).

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (2)$$

Where:

- x_i = investment in country i
- y_i = number or percentage of completed projects in country i

By integrating these metrics with qualitative assessments of permitting systems, certification mechanisms, and international partnerships, the study offers a multidimensional view of hydrogen market dynamics. The findings aim to inform policymakers, investors, and industry leaders about which regions are best positioned to shape the future of hydrogen.

Global Hydrogen Strategies and Targets

National Hydrogen Pledges, Investments, and Projects

Overview of national hydrogen strategies. National hydrogen strategies have emerged as a central component of decarbonization efforts across major economies, each reflecting distinct policy priorities, technological preferences, and timelines. Japan was among the first to launch a hydrogen strategy in 2017, envisioning a "hydrogen society" with a updated production targets of 3 million tonnes by 2030, 12 million by 2040, and 20 million by 2050 (GH2, 2025). South Korea's 2019 roadmap similarly emphasizes hydrogen mobility and energy diversification, aiming to supply 3.86 million tonnes annually by 2030 and 5.26 million tonnes by 2040, in addition to importing 1.96 million tonnes per year (Austrade, 2024). Within the Asia-Pacific region, Australia has announced its commitment to producing at least 15 million tonnes of hydrogen annually by 2050, supported by an interim export goal of 0.2 million tonnes of renewable hydrogen by 2030, signaling its intent to become a major supplier in the global hydrogen market (Baker McKenzie, 2024).

The European Union (EU) adopted its hydrogen strategy in 2020, with targets of 40–50 GW of electrolyzer capacity by 2030, with 85% expected to be powered by renewable energy sources (IRENA, 2024). This ambition is supported by substantial funding of approximately \$4.56 billion annually between 2021 and 2030, and reinforced by national strategies from member countries such as Germany, France, and

the Netherlands. Germany, for instance, aims to install 5 GW of electrolyzer capacity by 2030, while France has set an original target of 6.5 GW set in 2020 and then revised its target to 4.5 GW in 2025 (IEA, 2021; France Ministry of Economy, 2025).

The United States, through its 2023 strategy, has committed to producing 10 million metric tonnes (MMT) of clean hydrogen annually by 2030, scaling to 50 MMT by 2050 with an interim production goal of 20 MMT by 2040 (U.S. DOE, 2024). Canada's approach is technology-neutral, targeting 4 MMT by 2030 and 20 MMT by 2050, with a focus on reducing carbon intensity rather than specifying production methods (IRENA, 2024).

In the Middle East, the UAE's National Hydrogen Strategy 2050 outlines a phased production plan, starting with 1.4 MMT by 2031, 7.5 MMT by 2040, and ending with 14.9 MMT by 2050. This is supported by infrastructure development, including hydrogen clusters and transport corridors, and the deployment of the region's first solar-powered electrolyzer (UAE Government, 2024). Other Gulf Cooperation Council (GCC) countries have also outlined national hydrogen strategies. Oman, for example, plans to produce a minimum of 1 million metric tonnes (MMT) of renewable hydrogen annually by 2030, scaling up to 8.5 MMT by 2050 (International Energy Agency, 2024). Similarly, Saudi Arabia has set a target of generating 4 MMT of clean hydrogen by 2030, supported by large-scale projects such as the NEOM green hydrogen initiative (Baker McKenzie, 2024). These national targets are visually summarized in Fig. 1, which compares hydrogen production goals by country for 2030, 2040, and 2050.

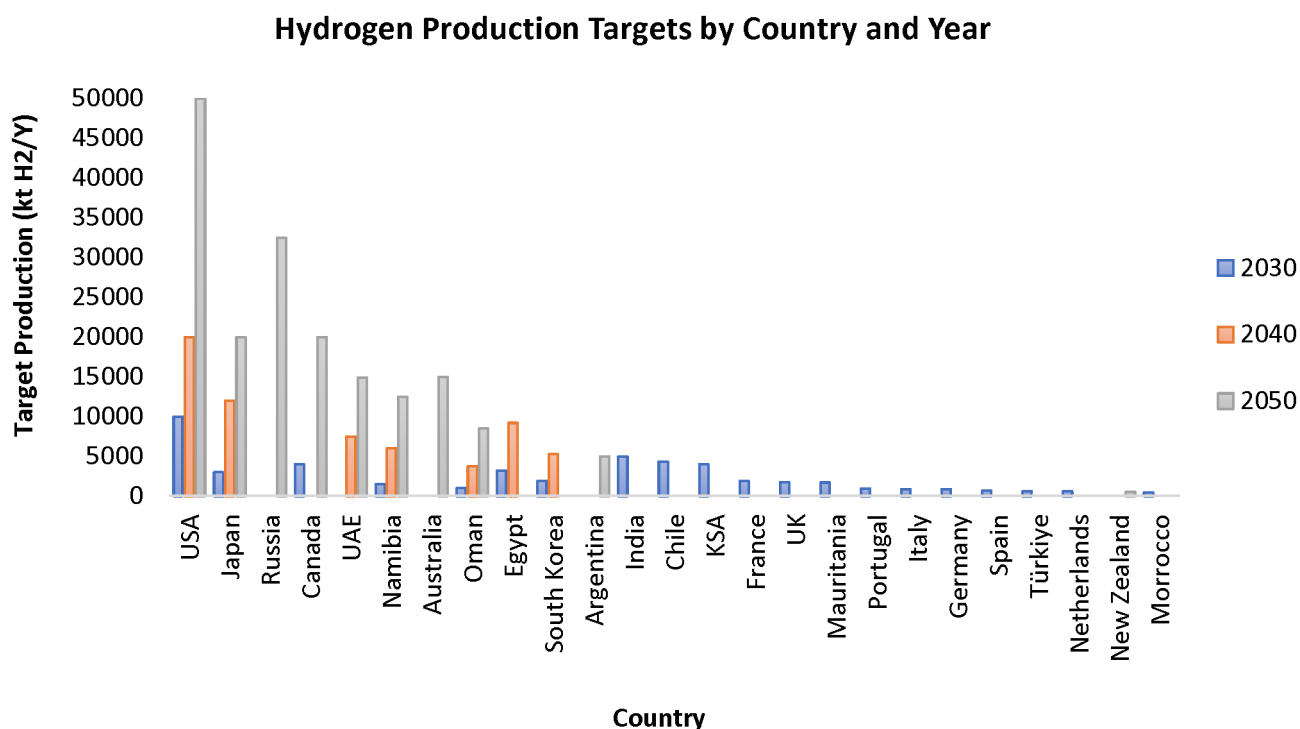


Figure 1—This chart presents hydrogen production targets (in kilotons per year) for selected countries, categorized by the years 2030, 2040, and 2050. It highlights national ambitions for scaling hydrogen output as part of global energy transition strategies.

In contrast, Africa's hydrogen ambitions are still in the early stages, with most projects concentrated in North African countries such as Morocco, Egypt, and Mauritania. Additionally, the Economic Community of West African States (ECOWAS), which includes 15 member states located in West Africa, has adopted a regional green hydrogen strategy. The framework sets a short-term target of 0.5 million tonnes of green hydrogen annually by 2030 and a long-term goal of 10 million tonnes by 2050 (Akinlabi et al., 2024).

These strategies collectively underscore the global momentum toward hydrogen as a key vector for energy transition, though they vary widely in scope, feasibility, and implementation readiness.

Comparative analysis of production targets. A comparative analysis of national hydrogen strategies has been conducted which reveals substantial variation in how quickly countries must scale their hydrogen production to meet stated goals. This section focuses on the time remaining to reach targets, the annual growth rates required to reach the targets, with using the annual current production volumes as a key indicator on the country's progress, all key indicators of strategic feasibility. Fig. 2 illustrates the forecasted annual growth rates required by countries to meet their hydrogen production targets, plotted against their respective timelines and categorized by feasibility level.

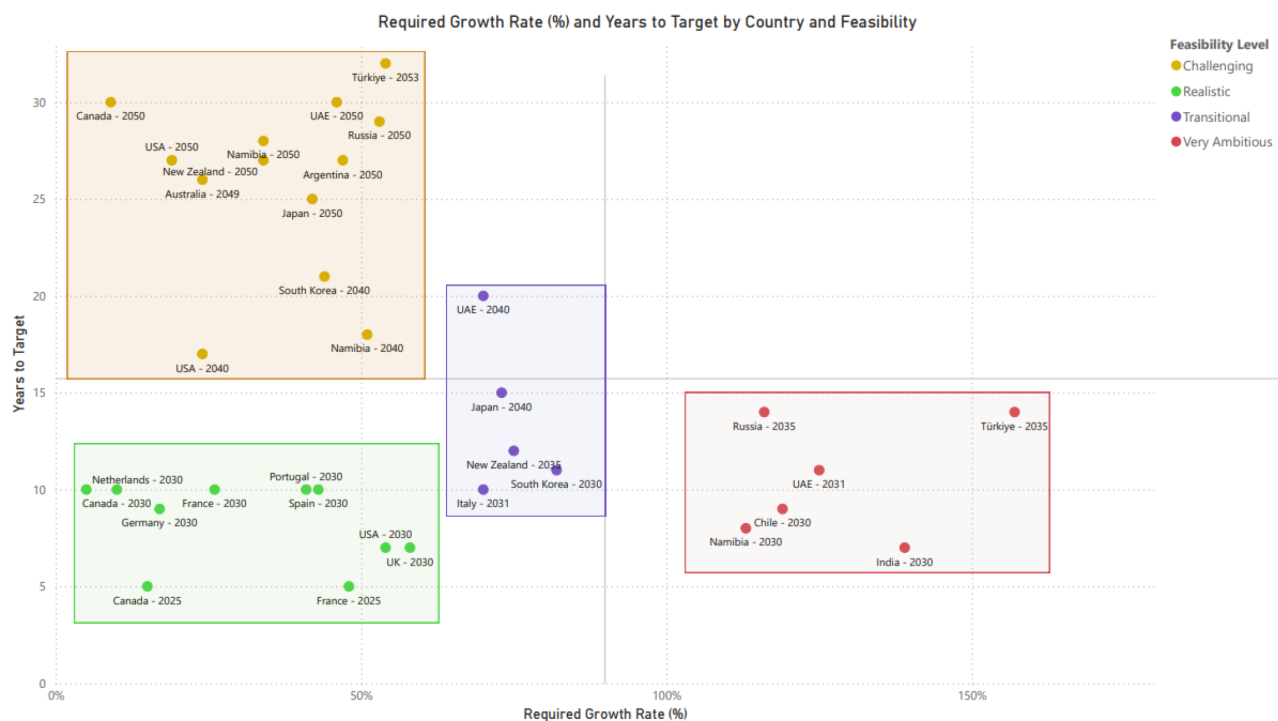


Figure 2—Projected annual growth rates required by countries to achieve their hydrogen production targets are plotted against their respective timelines. The visualization categorizes national strategies into three feasibility levels—Realistic (Green), Transitional (purple), Challenging (orange), and Very Ambitious (Red).

Countries such as Canada, Germany, and France fall within the realistic category, requiring less than 20% annual growth. These nations have between 5 and 10 years to reach their goals and currently produce between 0.11 and 1.48 MMT of hydrogen annually. Their required growth rates range from 9% to 15%, suggesting that their goals could be achievable with steady policy support and infrastructure expansion.

In contrast, countries like Japan, South Korea, and Namibia exhibit challenging trajectories, with growth rates between 20% and 50%. Japan, for example, has only 15 years to meet its 2040 milestone and must grow at 73% annually, despite producing just 0.0031 MMT currently. South Korea, with 11 years to target, requires 82% annual growth, starting from 0.0026 MMT. Namibia's 2030 goal demands 113% growth over 8 years, with current output at 0.0035 MMT. These figures reflect aggressive scaling needs from low baselines.

The most demanding category includes countries with very ambitious strategies, where annual growth rates exceed 100%. This group includes Chile, Türkiye, Russia, and India, among others. Chile's 2025 milestone requires 289% growth over just 4 years, starting from 0.0038 MMT. Türkiye's 2030 goal demands 247% growth, with current production at only 0.000009 MMT. Russia's 2024 export target requires over 1,000% annual growth, starting from 0.00015 MMT. This figure highlights the steep challenge of scaling from minimal production within short timeframes, which is quite unrealistic.

Overall, the feasibility breakdown shows that:

- **Realistic strategies** are characterized by growth rates below 80% and shorter timelines (≤ 16 years). These are considered achievable with current technologies and policy support, especially for countries with established production infrastructure.
- **Challenging strategies** involve moderate growth rates ($\leq 80\%$) but longer timelines (> 16 years). These require sustained scaling efforts and long-term investment, making them feasible but demanding.
- **Transitional strategies** represent a middle ground, with growth rates between 50–80% and timelines of 10–15 years. These are sensitive to policy, investment, and technological progress, and could shift toward either feasibility or difficulty depending on external factors.
- **Very Ambitious strategies** involve growth rates exceeding 100%, often paired with short timelines or low starting points. These are considered highly improbable without transformative changes in infrastructure, technology, and governance.

This analysis underscores that while hydrogen strategies are widespread and ambitious, their feasibility varies significantly. Without accelerated infrastructure development, policy reform, and international collaboration, over 40% of the reviewed targets risk remaining aspirational rather than achievable.

Investment Landscape and Project Development

Overview of Investment Distribution

The global hydrogen market is undergoing a transformative shift, marked by substantial but uneven investment across regions. The total announced investment amounts to approximately \$363.23 billion, reflecting a growing international consensus on hydrogen's role in achieving energy transition and climate goals. However, the scale, focus, and maturity of these investments vary significantly by region, as illustrated in Fig. 3.

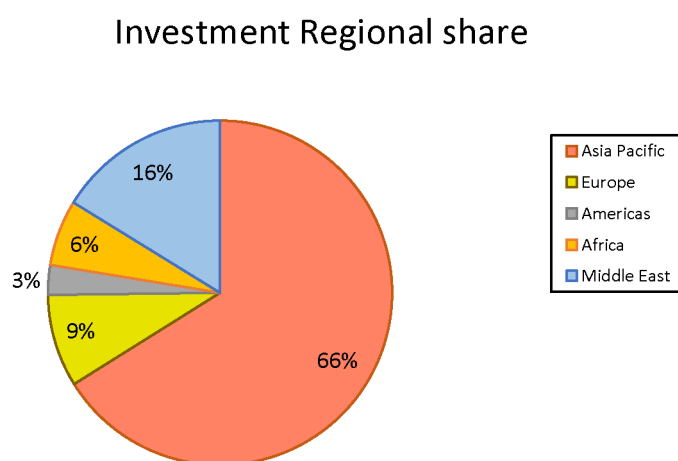


Figure 3—Regional distribution of investment shares.

In the Asia Pacific, countries are making bold financial commitments to position themselves as leaders in hydrogen innovation. Japan has pledged \$112.5 billion, the highest single-country investment globally, underscoring its ambition to build a comprehensive hydrogen economy (JETRO, 2025). South Korea has invested \$701.9 million to support its hydrogen vehicle industry and infrastructure (GH2, 2022), while India has committed \$ 2.31 billion to scale hydrogen production and integrate it into its national energy strategy (MNRE, 2023). Australia, through a bilateral agreement with Germany, has secured \$ 421 million

to develop green hydrogen supply chains (Baker McKenzie, 2025), and New Zealand has committed \$218 million through various government programs (MBIE, 2024).

The Middle East is emerging as a strategic hub for hydrogen development, leveraging its energy expertise and infrastructure. The United Arab Emirates (UAE) has announced \$1.63 billion in industrial hydrogen projects, including the country's first electrolyzer plant (FuelCellsWorks, 2023). Oman has attracted \$8.4 billion in investment, supported by major players such as Saudi Aramco and ACWA Power (Majan Council, 2024). Saudi Arabia is investing heavily, with \$5 billion allocated to the NEOM project and an additional \$10 billion through its Public Investment Fund (Air Products, 2020; Arabian Post, 2024).

In Africa, investment levels are elevated, driven by international partnerships and the continent's potential for renewable energy-based hydrogen production. Mauritania has announced up to \$80 billion in planned hydrogen projects, including a \$34 billion initiative involving Emirati, Egyptian, and German stakeholders (Argus Media, 2024; Taqa, 2024). Egypt has secured \$40 billion through seven major agreements (Egypt Today, 2024), and Morocco has approved six green hydrogen projects totaling \$32.8 billion (Hydrogen Insight, 2024). These investments reflect Africa's ambition to become a global exporter of green hydrogen.

Europe presents a diverse investment landscape, often tied to regulatory frameworks and technological leadership. Germany has committed \$20.5 billion through its Climate and Transformation Fund, with additional support from the IPCEI Hydrogen program (Hydrogen Insight, 2024). France has allocated \$ 2.1 billion to green hydrogen projects (IEA, 2024), while Italy has reserved an average value of \$21 billion for various hydrogen production technologies (Enagás, 2024). Portugal is investing around \$8 billion to develop a decarbonized hydrogen industry (Clifford Chance, 2020), and the Netherlands has introduced a \$2.4 billion subsidy program alongside a 4% mandate for industrial green hydrogen use (H2 View, 2024). While Russia has pledged \$125 million to develop its hydrogen industry (Hydrogen Insight, 2024) can be considered moderate compared to other European countries.

In the Americas, investment is more modest but strategically targeted. Canada has invested \$9.14 million in clean hydrogen projects, including funding for R&D and pilot plants (Natural Resources Canada, 2024). Chile has allocated \$50 million across six projects, supported by companies such as Enel Green Power and Air Liquide (Reuters, 2024). Argentina has launched an investment incentive regime exceeding \$200 million (Buenos Aires Herald, 2024).

This regional overview illustrates the varied strategic approaches to hydrogen investment. While some countries focus on large-scale infrastructure and export potential, others prioritize pilot projects, R&D, and regulatory alignment. The disparity in investment volumes reflects differences in national energy policies, resource availability, and the maturity of hydrogen ecosystems. As global momentum builds, these investments will play a critical role in shaping the future of clean energy.

Correlation Between Investment and Project Completion

The relationship between financial investment and project completion in the hydrogen sector is complex and varies significantly across countries. Using the Pearson correlation coefficient, this section evaluates whether higher investment levels are associated with increased project completion. A positive correlation ($r > 0$) suggests that as investment increases, project completion tends to rise, indicating efficient deployment. Conversely, a negative correlation ($r < 0$) implies that higher investment may not lead to more completed projects, potentially due to delays, complexity, or strategic long-term planning.

The analysis reveals several key patterns. Japan, despite being the top investor with \$112.5 billion, exhibits a strong negative correlation of -164.71, with only 19 completed projects as reported (IEA, 2024). This suggests that Japan's investments may be directed toward long-term infrastructure or face systemic delays. In contrast, countries such as South Korea ($r = 425.00$), Namibia ($r = 417.76$), and Portugal ($r = 288.76$) show extremely high positive correlations, though each has only one completed project (IEA, 2024). These outliers likely reflect disproportionately large investments in single projects, skewing the correlation.

Moderate correlations are observed in countries with more balanced investment and project completion profiles. For example, the UK ($r = 31.55$) has 14 completed projects (IEA, 2024), and Canada ($r = 42.78$) has 10, indicating a more proportional relationship. Germany, with 58 completed projects (IEA, 2024) and a correlation of 6.29, exemplifies efficient deployment of hydrogen investments. These findings underline that while investment is a critical enabler, project completion is influenced by additional factors such as regulatory efficiency, technological readiness, and strategic planning.

Hydrogen Deployment and Market Dynamics Across Regions

This analysis is based on the *Hydrogen Production and Infrastructure Projects Database* published by the International Energy Agency (IEA, 2024), which provides comprehensive data on global hydrogen production volumes, infrastructure projects, and development stages. The study further examines these insights in relation to investment trends and regional projections under the Net Zero Scenario, as outlined in the *Global Hydrogen Review 2024* (IEA, 2024). By integrating these datasets, the analysis aims to present an objective overview of the current state and strategic trajectory of hydrogen markets across regions and countries.

Regional Overview

The global hydrogen market exhibits significant regional disparities in production capacity, project volume, and investment intensity. The Americas lead in total hydrogen production, generating approximately 1980.6 kilometric tonnes of hydrogen per year (kMT H₂/year) from 324 projects. Europe, despite hosting the highest number of projects (863), produces only 790.8 kMT H₂/year, indicating the possibility of smaller-scale or pilot initiatives compared to the projects in the Americas. Asia Pacific, with 317 projects and 79.1 kMT H₂/year, reflects a predominance of early-stage developments. Africa and the Middle East, with 78 and 53 projects respectively, contribute minimally to the current global hydrogen output, 3.5 kMT H₂/year and 0.2 kMT H₂/year, despite Africa's relatively high project count (Consolidated Hydrogen Market Data, 2025).

Project development stages further illustrate regional maturity. Europe and Asia Pacific dominate in feasibility and concept phases, with Europe hosting 182 concept and 335 feasibility projects, and Asia Pacific 88 and 120 respectively. The Americas show a more mature pipeline, with 45 operational projects and 38 in final investment decision (FID) or construction stages. Fig. 4 visualizes these differences in project maturity across regions using a radar chart, highlighting the comparative advancement in each development stage. Investment patterns reveal strategic priorities: Asia Pacific leads with \$4.99 billion in announced investment, followed by the Middle East (\$1.23 billion), Africa (\$467 million), Europe (\$656 million), and the Americas (\$210 million). Notably, Africa's announced investments are disproportionately high relative to its production, suggesting speculative or long-term strategic interest. Asia Pacific currently leads the market with a 66.1% share, reflecting its active engagement in hydrogen development.

Hydrogen Project Maturity by Region

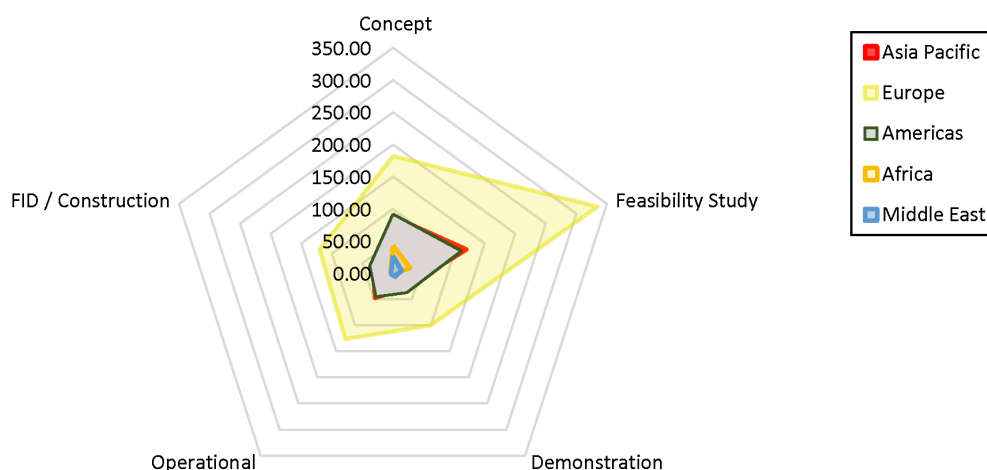


Figure 4—Radar chart comparing hydrogen project maturity across regions based on five development stages: Concept, Feasibility Study, Demo, Operational Projects, and FID/Construction.

The Net Zero Scenario, as defined by IEA, outlines a pathway for achieving net-zero greenhouse gas emissions globally by 2050. To align with this scenario, the IEA estimates that approximately \$ 50 billion in annual investment is required throughout this decade to scale hydrogen production and infrastructure to the necessary levels (IEA, 2024). This benchmark provides a reference point for evaluating regional investment commitments and projected market shares. Fig. 5 illustrates the contrast between current regional hydrogen investment shares and those projected under the Net Zero Scenario. Under the Net Zero Scenario, the Middle East commands a projected 94.3% regional share, underscoring its anticipated role as a future hydrogen hub. However, Europe and the Americas, while currently more mature in terms of operational projects and policy frameworks, show relatively modest shares under the Net Zero Scenario estimations 51% and 16%, respectively. This might suggest that despite their leadership in early deployment and regulatory clarity, their long-term production potential may be constrained by resource availability, cost structures, or strategic focus on hydrogen importation rather than large-scale domestic production.

Regional Hydrogen Investment Shares

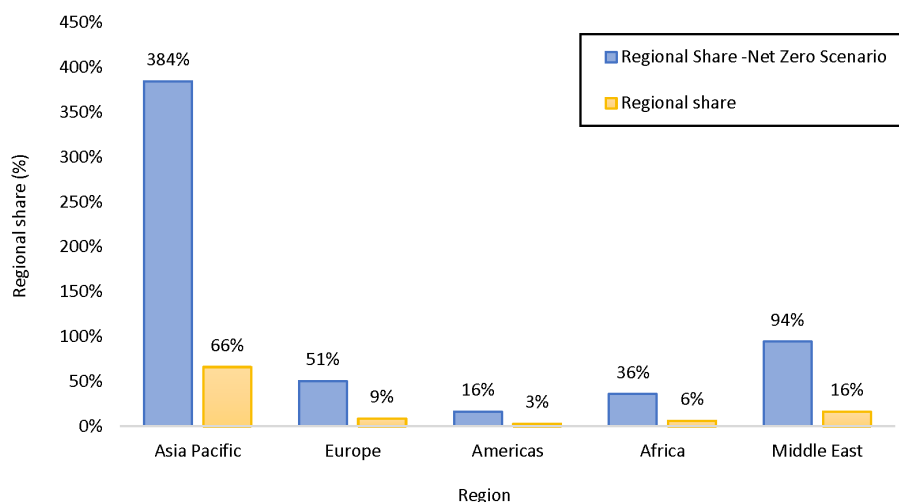


Figure 5—Comparison of Regional Hydrogen Investment Shares: Net Zero Scenario vs. Current Distribution

Hydrogen Demand Outlook

This section draws on findings from the study: *A Global Energy System Perspective on Hydrogen Trade: A Framework for the Market Color and Size Analysis* (Zhang, Wang, & Li, 2022), which models long-term hydrogen demand trajectories across global regions. According to the study, Asia Pacific is projected to be the most demanding region for low-carbon hydrogen by 2050, with its consumption curve dominating in both scale and duration. This projection aligns with the region's current investment momentum and policy frameworks and is consistent with this study's Net Zero Emissions by 2050 Scenario, which anticipates significant hydrogen uptake in Asia Pacific.

According to (Zhang, Wang, & Li, 2022), North America is expected to experience the fastest growth in hydrogen demand, particularly after 2030, driven by industrial uptake and policy incentives. Europe shows steady but moderate growth, supported by regulatory clarity and cross-border hydrogen strategies. Meanwhile, the (Zhang, Wang, & Li, 2022) study projects more modest demand increases for the Middle East and Africa. However, this finding appears to contradict the investment and strategic targets reported in the current dataset. The Middle East has announced over \$ 1.2 billion in hydrogen-related investments and is projected to hold a dominant share under the Net Zero Scenario, indicating ambitions to become a global hydrogen export hub. Similarly, Africa has committed \$ 467 million in investment despite its currently low production levels, suggesting future-oriented planning and international collaboration. These discrepancies highlight the importance of integrating both demand modelling and investment trajectories when assessing regional hydrogen market potential.

Economic Viability and Comparative Framework for Hydrogen Fuel Adoption

Economics of Hydrogen Production. Hydrogen production costs vary depending on the method and feedstock used. Green hydrogen via electrolysis, which is carbon-neutral (0 gCO₂eq/MJ), currently has a high production cost at \$6–\$8/kg H₂, primarily due to electricity input and infrastructure requirements.

Grey hydrogen from natural gas and brown hydrogen from coal gasification are significantly cheaper, ranging from \$2–\$6/kg H₂ and \$2–\$3/kg H₂, respectively. However, these methods are highly carbon-intensive, emitting > 12 kg CO₂/kg H₂ and >20 kg CO₂/kg H₂, respectively. Even with carbon capture and sequestration (CCS), the emissions remain substantial, and the cost increases to \$4–\$8/kg H₂ for grey hydrogen and \$4–\$5/kg H₂ for brown hydrogen.

This cost-emissions trade-off is central to evaluating hydrogen's economic viability. While fossil-based hydrogen remains cheaper, its environmental impact is significantly higher. Green hydrogen, though more expensive, offers a pathway to deep decarbonization if supported by policy incentives or low-cost renewable electricity.

Comparative Framework: Hydrogen vs. Displaced Fuels. To understand hydrogen's potential as a replacement fuel, it is useful to compare its carbon intensity and cost with conventional fossil fuels:

Table 1—Carbon Intensity and Production Costs of Hydrogen and Conventional Fossil Fuels

Fuel Type	Carbon Intensity	Cost Range
Green Hydrogen	0 gCO ₂ eq/MJ	\$6–\$8/kg H ₂
Grey Hydrogen	+12 kg CO ₂ /kg H ₂	\$2–\$6/kg H ₂
Brown Hydrogen	+20 kg CO ₂ /kg H ₂	\$2–\$3/kg H ₂
Natural Gas	56.1 kg CO ₂ e/GJ	\$0.5–\$1.7/kg Natural Gas
Diesel	70.27 kg CO ₂ e/GJ	\$1.21/kg Diesel
Coal	90.98 kg CO ₂ e/GJ	\$0.049/kg Coal

This comparison helps clarify hydrogen's role in replacing fossil fuels, especially in industries and transport where both cost and emissions matter. It shows which hydrogen types are clean enough and affordable enough to compete with fuels like diesel, coal, and natural gas. For example, green hydrogen offers zero emissions but at a higher cost, while grey and brown hydrogen are cheaper but significantly more polluting.

Supply and Cost Impacts Across Hydrogen Production Technologies. Hydrogen production costs are closely tied to the supply and pricing of key inputs, which vary significantly across technologies. For green hydrogen, produced via electrolysis, the process requires high-purity water (about 9 liters per kg H₂) and large amounts of electricity. The system includes electrolyzer stacks and balance-of-plant components such as cooling systems, rectifiers, water purification units, and hydrogen compressors, all contributing to high capital costs. Electricity is the dominant cost driver, accounting for up to 70% of operational expenses, and green hydrogen only becomes competitive when renewable electricity prices fall below \$20–\$30/MWh.

Blue hydrogen (estimated at \$2.00–\$3.50/kg H₂) is produced via steam methane reforming (SMR) with carbon capture and storage (CCS). It relies on natural gas and additional infrastructure for CO₂ compression, transport, and geological sequestration. CCS can add 20–40% to capital costs, especially in regions lacking established storage infrastructure, because electricity makes up the majority of its production cost.

Gray hydrogen is also produced via SMR but without CCS. It has the lowest capital costs, as existing refinery and industrial facilities can be leveraged with minimal modifications. However, it is increasingly vulnerable to carbon pricing, which could raise its cost above \$4/kg H₂ at \$200/ton CO₂.

Overall, the levelized cost of hydrogen (LCOH) to the consumer is shaped by the availability and cost of electricity, water, and natural gas, as well as policy incentives and infrastructure readiness. Green hydrogen requires the most favorable conditions to achieve cost parity, while blue hydrogen serves as a transitional option and grey hydrogen faces long-term regulatory constraints.

Regulatory Frameworks and Deployment Challenges

The success of hydrogen market deployment is increasingly tied to the presence of comprehensive and coherent regulatory systems. Countries with clear federal legislation, including Brazil, France, the USA, South Korea, Egypt, the European Union, and the UAE, have established harmonized laws that define hydrogen classifications, environmental standards, and safety protocols ([Global Compliance News, 2024](#); [Reed Smith, 2022](#); [U.S. Congress, 2021](#); [KLRI, 2021](#); [Baker McKenzie, 2024](#); [European Commission, 2024](#)). These frameworks reduce legal uncertainty, facilitate investment decisions, and accelerate project approvals. For example, the USA's Infrastructure Investment and Jobs Act and the EU's hydrogen and gas decarbonization package provide robust legal foundations that support long-term hydrogen development ([U.S. Congress, 2021](#)). [Fig. 6](#) illustrates the presence of hydrogen policy frameworks across countries, categorized by six key factors. Each blue bar indicates active implementation of the respective policy in the corresponding country.



Figure 6—Gantt chart of the present hydrogen policy frameworks across countries, categorized by six key factors.

A critical enabler of project efficiency is the implementation of one-stop shop permitting systems, which consolidate multiple administrative procedures into a single regulatory interface. This model has reported to be currently operational in South Korea, Shanghai (China), the Netherlands, UAE, Oman, and Germany (GH2, 2023; Verticer, 2023; GigCI, 2023; HKTDC, 2025; BMWK, 2024). These systems significantly reduce bureaucratic delays and enhance project execution timelines.

Despite these advancements, explicit hydrogen policies and national roadmaps remain uneven across regions. As discussed in section 3.1.1. of this paper, many countries from different regions have published formal strategies that outline production targets, sectoral integration, and timelines. These roadmaps provide clarity to investors and signal long-term government commitment. In contrast, countries lacking such

frameworks face fragmented efforts and limited investor confidence, which can hinder project financing and execution.

International collaboration and trade agreements are also pivotal in shaping hydrogen market dynamics. Bilateral and multilateral partnerships, spanning Australia, Austria, Belgium, Canada, Chile, China, Egypt, France, Germany, India, Japan, Kenya, Namibia, the Netherlands, New Zealand, Saudi Arabia, South Korea, the UAE, the UK, and the USA, facilitate hydrogen trade, technology transfer, and joint infrastructure development (Weltenergierat, 2021; World Energy Council, 2021). These agreements support cross-border hydrogen flows and help harmonize technical standards, which are essential for building global supply chains.

Another important aspect is the regional presence of hydrogen certification mechanisms which reflects growing regulatory sophistication. These systems verify the origin and carbon intensity of hydrogen, enabling classification into different categories. Certification is essential for compliance with environmental standards and eligibility for subsidies. Europe leads in this area with CertifHy and RED II, while other countries have adopted national schemes such as TÜV SÜD in Germany, AFHYPAC in France, Vergogas in the Netherlands, Certified Energy in New Zealand, EEGO in Portugal and Smart Energy Council in Australia. Emerging frameworks are also underway in the UAE and Morocco, indicating a broader regional commitment to transparency and sustainability (Cheng et.al, 2022).

However, permitting constraints remain a significant barrier in several major economies. The UK, EU, USA, India, and Australia face challenges such as fragmented jurisdictional oversight, lengthy environmental assessments, and unclear land use policies. These constraints delay project approvals and undermine investor confidence. For instance, local opposition and planning complexities in Australia’s Kalbarri region led to the rejection of a green hydrogen proposal, underscoring the need for inclusive and streamlined regulatory processes (UK Government, 2023; GH2, 2023; Energy News, 2024; Cedigaz, 2025; WFES, 2025; ABC News, 2024).

Comparative Regional Analysis and Recommendations

The summarizing matrix, shown in Table 2, offers a comprehensive and color-coded snapshot of hydrogen market readiness across five global regions. While disparities in performance are evident, the analysis reveals not only current gaps but also significant opportunities for strategic advancement. Each region is navigating its own trajectory, shaped by policy maturity, investment dynamics, and institutional capacity.

Table 2—Summarizing matrix of the regional hydrogen production targets, investments, and regulatory conditions across global regions. Target feasibility coloring scheme was used: Green (Realistic), Orange (Challenging), and Red (Very Ambitious).

Region	Total Production (kt H2/y)	Target Production (kt H2/y)	Target Completion (%)	Required Growth Rate (%)	Country Announced Investment (\$)	Project Completion Rate	Correlation (Investment vs Completion)	Regulatory Conditions
Asia Pacific	79.1	31866.3	0.2%	>50%	> 4 Billion	Few completed; many in planning	Low correlation; high pledges, low delivery	Mixed; strong in Japan, weaker in others
Europe	790.8	63763.5	1.2%	20-30 %	~ 600 Million	Many completed; strong execution	High correlation	Mature, integrated EU policy
Americas	1980.6	11525.0	17.2%	20-40%	~ 200 Million	Several completed (USA, Canada)	Moderate correlation	Strong in USA/Canada but with delays; Fragmented elsewhere
Africa	3.5	9764.6	0.0%	>70%	~ 500 Million	Minimal completions	Low correlation	Early-stage; limited regulatory clarity
Middle East	0.2	77687.5	0.0%	>60%	~ 1.2 Billion	Very few completed	Low correlation	Emerging frameworks; improving

The Americas currently lead in execution, with the highest target completion rate and strong project delivery. This reflects a pragmatic alignment between ambition and infrastructure, particularly in the USA and Canada. However, the region's regulatory environment, while functional, is beginning to show signs of fragmentation, especially in permitting and certification, which could slow future scalability if not addressed.

Europe continues to benefit from mature and integrated policy frameworks, which have historically positioned it as a global leader in hydrogen strategy. Yet, recent developments suggest that regulatory complexity and permitting delays are becoming bottlenecks. Without reform, Europe risks losing its competitive edge despite its strong institutional foundation.

In contrast, the Middle East is emerging as a dynamic and increasingly coordinated player. While current production and completion rates remain modest, the region's visionary targets, substantial investment, and improving regulatory frameworks signal a clear upward trajectory. Countries like the UAE and Oman are actively developing certification systems and streamlining permitting, laying the groundwork for accelerated deployment. With continued regional cooperation and integration into global hydrogen markets, the Middle East is poised to become a major exporter and innovation hub.

Africa, often viewed through the lens of feasibility constraints, is showing early signs of strategic alignment. Though execution remains low, the continent's abundant renewable resources, growing investment interest, and increasing participation in international hydrogen partnerships offer a promising foundation. With targeted support in policy development, infrastructure planning, and technology access, Africa could evolve into a key supplier of green hydrogen, especially to Europe and Asia.

Asia Pacific presents a mixed picture. The region boasts the highest investment and some of the most ambitious targets, yet delivery remains uneven. Countries like Japan and Australia are advancing rapidly, while others face regulatory and institutional hurdles. Harmonizing regional strategies and improving investment-to-execution efficiency will be critical to unlocking Asia Pacific's full potential.

Strategic Recommendations

- Accelerate Regulatory Reform in Mature Markets: Europe and the US must streamline permitting and certification to maintain leadership and avoid stagnation.
- Support Emerging Hydrogen Hubs: The Middle East and Africa should be prioritized for international collaboration, capacity building, and infrastructure investment to harness their strategic potential.
- Promote Regional Integration: Shared infrastructure, harmonized standards, and cross-border trade agreements can enhance scalability and reduce fragmentation, especially in Asia Pacific and Africa.
- Balance Ambition with Feasibility: Regions should align targets with realistic growth trajectories, ensuring that policy ambition is matched by institutional and technical capacity.
- Leverage Renewable Resource Advantages: Africa and the Middle East, with vast solar and wind potential, should focus on green hydrogen pathways that align with global decarbonization goals.
- Future studies should deepen regional analysis by evaluating policy-technology alignment, investment effectiveness, and cross-border collaboration to guide more targeted and scalable hydrogen strategies.

Conclusion

This study set out to answer a pivotal question: *Who is most likely to lead the global hydrogen market in the coming decades?* Through a comparative analysis of national strategies, investment patterns, project maturity, and regulatory frameworks, the findings reveal a complex and uneven global landscape.

Regions such as Europe and the Americas currently lead in terms of project maturity and operational capacity, supported by robust regulatory frameworks and early strategic adoption. Asia Pacific, while

exhibiting the highest investment volumes, shows a mixed picture in terms of project execution, with countries like Japan and South Korea facing steep growth requirements from low baselines. The Middle East and Africa, though still in early stages of deployment, are emerging as ambitious players, with visionary targets and growing international partnerships positioning them as future hydrogen exporters.

The use of the Growth Rate of Hydrogen Capacity as a predictive indicator revealed that many countries—particularly in Asia, Africa, and the Middle East—face exponential scaling challenges to meet their 2030 and 2050 targets. Meanwhile, the correlation between investment and project completion highlighted that financial commitment alone does not guarantee progress; regulatory efficiency, permitting systems, and institutional capacity are equally critical.

Ultimately, no single region currently dominates across all dimensions. However, Europe and select countries in the Americas appear best positioned to maintain leadership in the short term, while the Middle East and Asia Pacific may emerge as dominant players in the long term—provided they can translate ambition into execution. Africa, with its renewable resource potential and growing international support, holds promise as a strategic supplier, particularly to Europe and Asia.

As the hydrogen economy evolves, leadership will depend not only on technological and financial capacity but also on the ability to align policy, infrastructure, and international cooperation. The global race for hydrogen dominance is underway—and its outcome will shape the future of clean energy.

References

- ABC News. (2024). Green hydrogen push rejected by WA town Kalbarri. <https://www.abc.net.au/news/2024-04-30/green-hydrogen-push-rejected-by-wa-town-kalbarri/103741566>
- Air Products. (2020). Air Products, ACWA Power and NEOM sign agreement for \$5 billion green ammonia production facility. <https://www.airproducts.com/company/news-center/2020/07/0707-air-products-agreement-for-green-ammonia-production-facility-for-export-to-hydrogen-market>
- Akinlabi, B., Akinlabi, S., & Madushele, N. (2024). Green hydrogen development in West Africa: Opportunities, challenges and policy recommendations. *International Journal of Hydrogen Energy*. <https://www.sciencedirect.com/science/article/pii/S0360319924036863>
- Arabian Post. (2024). Saudi Arabia pushes forward with \$10bn green hydrogen investment. <https://thearabianpost.com/saudi-arabia-pushes-forward-with-10bn-green-hydrogen-investment/>
- Argus Media. (2024). Mauritania's planned H2 projects total \$80GW – minister. <https://www.argusmedia.com/en/news-and-insights/latest-market-news/2447263-mauritania-s-planned-h2-projects-total-80gw-minister>
- Argus Media. (2024, July 10). Africa's H₂ project development lags behind: Report. <https://www.argusmedia.com/en/news-and-insights/latest-market-news/2552108-africa-s-h2-project-development-lags-behind-report>
- Austrade. (2024). Korean Hydrogen Market Report. https://www.intralinkgroup.com/getmedia/8ba563f7-08f6-43de-ab4a-4467c7e8b3bd/Austrade-Korean-Hydrogen-Market_Report
- Baker McKenzie. (2024). Egypt: New incentives for green hydrogen and derivatives projects. <https://insightplus.bakermckenzie.com/bm/projects/egypt-new-incentives-for-green-hydrogen-and-derivatives-projects>
- Baker McKenzie. (2024). Hydrogen developments in Australia and Saudi Arabia. <https://resourcehub.bakermckenzie.com/en/resources/hydrogen-heat-map>
- Baker McKenzie. (2025). Hydrogen developments in Australia. <http://resourcehub.bakermckenzie.com/en/resources/hydrogen-heat-map/asia-pacific/australia/topics/hydrogen-developments>
- Benesty, J., Chen, J., Huang, Y., & Cohen, I. (2009). Pearson correlation coefficient. In *Noise reduction in speech processing* (Vol. 2, pp. 1–4). Springer. https://doi.org/10.1007/978-3-642-00296-0_5
- BMWK. (2024). Hydrogen home. <https://www.bundeswirtschaftsministerium.de/Redaktion/EN/Hydrogen/home.html>
- Buenos Aires Herald. (2024). Understanding Argentina's new large investment regime. <https://buenosairesherald.com/business/understanding-argentinas-new-large-investment-regime>
- Cedigaz. (2025). Global hydrogen market 2025 assessment. <https://www.cedigaz.org/global-hydrogen-market-2025-assessment>
- Cheng, C., Zhang, Y., & Li, H. (2022). Hydrogen certification mechanisms: A review of global practices. *Sustainability*, 14(3), 1930. <https://www.mdpi.com/2071-1050/14/3/1930>
- Clifford Chance. (2020). Focus on hydrogen: Portugal's clean energy plan gathers pace. <https://www.cliffordchance.com/content/dam/cliffordchance/briefings/2020/11/focus-on-hydrogen-portugals-clean-energy-plan-gathers-pace.pdf>

- Climate Change Laws of the World. (2021). Development of hydrogen energy in the Russian Federation until 2024. https://climate-laws.org/documents/development-of-hydrogen-energy-in-the-russian-federation-until-2024_10d2
- Corporate Finance Institute. (n.d.). What is CAGR (Compound Annual Growth Rate)? Retrieved July 29, 2025, from <https://corporatefinanceinstitute.com/resources/valuation/what-is-cagr/>
- Curcio, E. (2025). Techno-economic analysis of hydrogen production: Costs, policies, and scalability in the transition to net-zero [Preprint]. arXiv. <https://doi.org/10.48550/arXiv.2502.12211>
- Egypt Today. (2024). Egypt inks 7 agreements with \$40 bln investments in hydrogen projects. <https://www.egypttoday.com/Article/3/130681/Egypt-inks-7-agreements-with-40-bln-investments-in-hydrogen>
- Enagás. (2024). Italy renewable hydrogen. <https://goodnewenergy.enagas.es/en/innovative/italy-renewable-hydrogen/>
- Energy News. (2024). European hydrogen projects face challenges. <https://energynews.biz/european-hydrogen-projects-face-challenges/>
- European Commission. (2024). Hydrogen and gas decarbonisation package. https://energy.ec.europa.eu/topics/markets-and-consumers/hydrogen-and-decarbonised-gas-market_en
- France Ministry of Economy. (2025). Stratégie nationale hydrogène décarboné 2025. https://www.economie.gouv.fr/files/2025-04/strategie_nationale_hydrogene_decarbone_2025.pdf
- FuelCellsWorks. (2023). UAE invests \$1.63 billion in industrial projects, including the country's first hydrogen electrolyzer plant. <https://fuelcellsworks.com/news/uae-invests-1-63-billion-in-industrial-projects-including-the-countrys-first-hydrogen-electrolyzer-plant>
- GH2. (2022). South Korea country profile. <https://gh2.org/countries/south-korea>
- GH2. (2023). PCC Report. https://gh2.org/sites/default/files/2023-09/GH2_PCC%20Report_18092023.pdf
- GH2. (2025). Japan Hydrogen Strategy (English Version). <https://gh2.org/sites/default/files/2025-03/WP%20Japan%20%28English%20Version%29.pdf>
- GigCl. (2023). Hydrogen energy. <https://gigcl.ae/hydrogen-energy/>
- Global Compliance News. (2024). Brazil: The legal framework for low-carbon hydrogen. <https://www.globalcompliancencews.com/2024/09/04/brazil-the-legal-framework-for-low-carbon-hydrogen/>
- H2 View. (2024). Netherlands commits €2.1bn to green hydrogen production and introduces 4% industry mandate. <https://www.h2-view.com/story/netherlands-commits-e2-1bn-to-green-hydrogen-production-and-introduces-4-industry-mandate/2125126.article>
- HKTDC. (2025). Hydrogen permitting in China. <https://research.hktdc.com/en/article/MjAyNTI5OTY0OQ>
- Hydrogen Insight. (2024). German government to spend €18.6bn on hydrogen from its Climate and Transformation Fund in 2024–27. <https://www.hydrogeninsight.com/policy/german-government-to-spend-18-6bn-on-hydrogen-from-its-climate-and-transformation-fund-in-2024-27/2-1-1499546>
- Hydrogen Insight. (2024). Morocco approves six green hydrogen projects with a total investment value of \$32.8bn. <https://www.hydrogeninsight.com/production/morocco-approves-six-green-hydrogen-projects-with-a-total-investment-value-of-32-8bn/2-1-1789524>
- IEA. (2021). Package for the Future: Hydrogen Strategy – Germany. <https://www.iea.org/policies/11561-package-for-the-future-hydrogen-strategy>
- IEA. (2024a). Hydrogen Production and Infrastructure Projects Database. International Energy Agency. Retrieved from <https://www.iea.org/data-and-statistics/data-product/hydrogen-production-and-infrastructure-projects-database>
- IEA. (2024b). Global Hydrogen Review 2024. International Energy Agency. Retrieved from <https://www.iea.org/reports/global-hydrogen-review-2024>
- International Energy Agency (IEA). (2024). Hydrogen production and infrastructure projects database. <https://www.iea.org/data-and-statistics/data-product/hydrogen-production-and-infrastructure-projects-database>
- International Energy Agency. (2024). Oman's renewable hydrogen potential. <https://www.iea.org/news/oman-s-huge-renewable-hydrogen-potential-can-bring-multiple-benefits-in-its-journey-to-net-zero-emissions>
- IRENA. (2024). Green Hydrogen Strategy Design. https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2024/Jul/IRENA_Green_hydrogen_strategy_design_2024.pdf
- JETRO. (2025). Japan's hydrogen revolution. <https://www.jetro.go.jp/en/invest/insights/japan-insight/japans-hydrogen-revolution.html>
- KLRI. (2021). Hydrogen law viewer. https://elaw.klri.re.kr/eng_mobile/viewer.do?hseq=60917&type=sogan&key=13
- Majan Council. (2024). Hydrom signs \$20 billion deals for green hydrogen in Oman. <https://www.majancouncil.org/18374/>
- MBIE. (2024). Interim hydrogen roadmap. <https://www.mbie.govt.nz/dmsdocument/26911-interim-hydrogen-roadmap-pdf>
- MNRE. (2023). Hydrogen overview. <https://mnre.gov.in/en/hydrogen-overview/>

- Natural Resources Canada. (2024). Minister Wilkinson announces \$125 million to support clean hydrogen development in Canada. <https://www.canada.ca/en/natural-resources-canada/news/2024/08/minister-wilkinson-announces-125-million-to-support-clean-hydrogen-development-in-canada.html>
- NSWS. (2023). National Green Hydrogen Mission – India. [https://www.nsws.gov.in/s3fs/2023-10/National%20Green%20Hydrogen%20Mission%20\(1\).pdf](https://www.nsws.gov.in/s3fs/2023-10/National%20Green%20Hydrogen%20Mission%20(1).pdf)
- Reed Smith. (2022). Hydrogen regulations by jurisdiction. <https://www.reedsmith.com/en/perspectives/energy-transition/2022/06/hydrogen-regulations-by-jurisdiction-and-changing-transmission-systems>
- Reuters. (2024). Chile leads Latin American push for clean hydrogen. <https://www.reuters.com/business/energy/chile-leads-latin-american-push-clean-hydrogen-2024-10-22/>
- SGH2 Energy. (n.d.). Economics. Retrieved August 23, 2025, from <https://www.sgh2energy.com/economics>
- Statista. (2025, August). Monthly retail prices for diesel fuel in the United States from January 2020 to July 2025 (in U.S. dollars per gallon) [Graph]. <https://www.statista.com/statistics/204169/retail-prices-of-diesel-fuel-in-the-united-states-since-2009/>
- Taq. (2024). Mauritania: Signing a memorandum of understanding to establish a green hydrogen project worth \$34 billion. <https://taqas.net/en/mauritania-signing-a-memorandum-of-understanding-to-establish-a-green-hydrogen-project-worth-34-billion/>
- U.S. Congress. (2021). Infrastructure Investment and Jobs Act. <https://www.congress.gov/bill/117th-congress/house-bill/3684>
- U.S. Department of Energy. (2024). National Hydrogen Strategy Remarks. <https://www.energy.gov/sites/default/files/2024-02/doc-h2-program-national-strategy-remarks-jan2024.pdf>
- U.S. Energy Information Administration. (n.d.). Voluntary reporting of greenhouse gases program: Fuel emission coefficients. Retrieved August 23, 2025, from https://www.eia.gov/environment/emissions/co2_vol_mass.php
- UK Government. (2023). Hydrogen projects planning barriers and solutions. <https://assets.publishing.service.gov.uk/media/657a2a92095987000d95e086/hydrogen-projects-planning-barriers-and-solutions-report.pdf>
- Verticer. (2023). About Verticer. <https://verticer.eu/en/about-verticer/>
- Weltenergierat. (2021). International hydrogen strategies. <https://www.weltenergierat.de/publikationen/studien/international-hydrogen-strategies/?cn-reloaded=1>
- WFES. (2025). Green Hydrogen and Future Fuels Report. <https://www.worldfutureenergysummit.com/content/dam/sitebuilder/rxae/worldfutureenergysummit/docs/WFES-ADSW-Report-2025-Green-Hydrogen-and-Future-Fuels.pdf.coredownload.504138533.pdf>
- World Energy Council. (2021). National Hydrogen Strategies. https://www.worldenergy.org/assets/downloads/Working_Paper_-_National_Hydrogen_Strategies_-_September_2021.pdf
- Zhang, Y., Wang, J., & Li, K. (2022). A review on data-driven approaches for energy system modeling and optimization. *Applied Energy*, **310**, 120267. <https://doi.org/10.1016/j.apenergy.2022.120267>